

MANAGEMENT OF MINOR FRACTURES, SPRAINS AND STRAINS OF THE UPPER LIMB

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CONTENTS

Introduction.....	3
Red Flags	3
Finger Sprains.....	4
Fractures	5
Finger Fractures	5
Metacarpal Fractures	6
Carpal Bone Fractures.....	6
Distal Radius Fractures.....	7
Follow-ups for Fractures	7
Adhesive Capsulitis.....	8
Common Upper Limb Tendinopathy	9
Rotator Cuff Tendinopathy	10
Epicondylitis.....	11
De Quervain's Tenosynovitis	12
Carpal Tunnel Syndrome	14
Trigger Finger	16
Reference	18

INTRODUCTION

- **Strain:** A tearing injury to muscle fibres from excessive tension or overuse
- **Sprain:** A tearing injury to one or more ligament of a joint when joint is forced beyond the limits of its normal plane of motion
- **Fracture:** A disruption in bone tissue from force exceeding the strength of the bone, repetitive stress or an invasive process
- Principles of acute management for minor strains, sprains and fractures
 - **P:** Protect – from further damage by using a support or splint (if required)
 - **R:** Rest – and avoid excessive walking or weight bearing on injured side
 - **I:** Ice – for 15 minutes every 2-3 hours (do not apply ice directly to skin)
 - **C:** Compress – with elastic bandage (do not apply when sleeping)
 - **E:** Elevate – and support to reduce swelling and bruising
- Avoid using ice beyond the first 24 hours of injury as it may potentially impair the healing response by inhibiting inflammation
- For strains and sprains, avoid prolonged rest which can cause stiffness and weaken tissues. When pain has improved, gentle range of movement exercises need to be encouraged
- When evaluating an injured extremity, always evaluate the joint above and the joint below the site of injury as indirect forces may have affected these sites as well
- Refer **ALL** injuries with **RED FLAGS** to the Emergency Department for further management, including situations where backslab / casts for immobilization are indicated

RED FLAGS

- Unstable fractures requiring backslab / casts for immobilization, or intra-articular fractures
- Fractures involving weight bearing bones or long bones
 - **Possible exception:** small avulsion / chip fractures without significant pain and not involving a joint
- Open fractures or fractures with significant soft tissue injuries (severe nail bed injuries, crush injuries *etc.*)
- Acute dislocations
- Any injury with neurovascular compromise

- Fracture lines may not be always visible / obvious in acute fractures
 - If high clinical suspicion (e.g., severe pain, significant swelling, immobility, functional impairment), patient should be managed as for a possible fracture
 - Otherwise, patient can be reviewed and radiographs can be repeated in 1 – 2 weeks after immobilization

FINGER SPRAINS

- Suggested investigation: XR Fingers AP and Lateral views
- Volar (Palmar side) plate sprain
 - From hyperextension injuries, pain usually over PIPJ or DIPJ
 - Can be treated with a buddy splint with an adjacent finger for 1 – 2 weeks
 - If the presence of avulsion fractures
 - Finger should preferably be splinted (Fig 1)
 - Refer to hand surgery direct access
- Collateral ligament injury
 - Majority can be managed conservatively with buddy splint and early mobilization (within 2 weeks)
 - **Except for** ulnar collateral ligament injuries of the 1st MCP joint
 - From hyperextension or hyperabduction
 - Pain, swelling, ecchymosis over thenar eminence + instability of 1st MCP joint
 - Refer to hand surgery direct access
- Traumatic flexor tendon avulsion injury of the FDP from the distal phalanx insertion (Jersey finger)
 - Inability to actively flex DIPJ
 - Suggested investigation: XR finger AP and Lateral views
 - The finger preferably should be splinted with PIPJ and DIPJ in slight flexion (Fig. 1)
 - **ALL** jersey finger injuries should be referred to the hand surgeon urgently (within 1 week preferably)



FRACTURES

FINGER FRACTURES

- Suggested investigation: XR Fingers AP and Lateral views
- Distal phalanx fractures
 - Tuft fractures, extensor terminal tendon insertion tear $\pm$ avulsion fractures (Mallet finger) can be managed conservatively
 - Mallet finger requires extension splint (Fig. 2a and Fig. 2b)
 - Refer Mallet finger injuries to hand surgery direct access
 - Refer acutely displaced, shortened, or angulated fractures to A&E
- Middle phalanx fractures
 - Commonly associated with tendon / soft tissue injuries leading to joint immobility
 - Need to be assessed for rotation, shortening, or angulation
 - Immobilise in a gutter splint (Fig 1)
 - Refer to hand surgery direct access
 - Refer acutely displaced, shortened, or angulated fractures to A&E as they may require M&R to improve alignment.
- Proximal phalanx fractures
 - Often unstable due to forces exerting on the fracture fragments
 - Refer to A&E for immobilization



Figure 2a



Figure 2b (available in NUP)

METACARPAL FRACTURES

- Suggested investigation: XR Hand PA and Oblique views
- First metacarpal
 - Intra-articular (Bennett's & Rolando's fractures): high likelihood of requiring surgical fixation
 - Extra-articular: can be managed conservatively with a long thumb spica splint or surgically
- 2nd to 5th metacarpal
 - Need to be accessed for angulation and scissoring of the fingers
 - Non-displaced fractures will require splint immobilisation
- Refer acute metacarpal fractures to the emergency department for reduction or immobilization
 - Small avulsion / chip fractures without significant pain may not require ED referral but early hand surgery review (direct access)
 - Review with senior doctors if unsure of the disposition

CARPAL BONE FRACTURES

- Suggested investigation: XR Wrist Joint AP and Lateral views
- Generally, all carpal bone fractures require immobilization with a backslab before a HRM review regarding the need for surgery
- Refer acute carpal bone fractures to the emergency department for immobilization
 - Small avulsion / chip fractures without significant pain may not require ED referral but early hand surgery review (direct access)
 - Review with senior doctors if unsure of the disposition plan
- Special attention to **Scaphoid fracture**
 - Suggested investigation: XR Scaphoid view
 - A radiograph taken soon after the injury may not reveal a fracture
 - Always ask for a scaphoid view if looking for a scaphoid fracture
 - Prevalence of occult fracture with trauma-related wrist pain but negative plain radiograph is around 25%

- If clinical suspicion is high (snuffbox tenderness *etc.*), should assume one is present until definite proof otherwise is obtained
- Requires immobilization in a thumb spica slab

DISTAL RADIUS FRACTURES

- Suggested investigation: XR Wrist Joint AP and Lateral views
- Colles' fracture: extra-articular and dorsal angulation of distal radius fragment
- Smith's fracture: extra-articular and volar angulation of distal radius fragment
- Requires splint / arm cast and early HRM review
- Refer acute distal radius fractures to the emergency department for immobilization (reduction if required).
 - Small avulsion / chip fractures without significant pain may not require ED referral but early HRM review (direct access)
 - Should suggest other forms of immobilization (e.g., a firm wrist guard)
 - Review with senior doctors if unsure of the disposition plan

FOLLOW-UPS FOR FRACTURES

- Minor fractures without the need to be referred to a specialist should be followed up closely in primary care with repeated X-rays to document healing
- Follow up with acute fractures within 1-2 weeks of initial injury to:
 - Repeat X-ray to ensure no significant displacement
 - Ensure adequate pain control
- The typical healing duration of limb fractures is usually 8-12 weeks
- Consider specialist referral in the presence of:
 - Worsening or persistent pain, swelling, or loss of function/disability due to the fracture
 - Malunion, Nonunion, or delayed union of the fracture site
 - Injuries where injury claim/compensation or legal input may be required

ADHESIVE CAPSULITIS

Introduction:

- Common shoulder condition characterized by pain, stiffness, and loss of function of the affected shoulder
- It primarily manifests as an idiopathic condition with increased prevalence in patients with hypothyroidism, diabetes, and women between 40 and 60 years of age
- While it commonly resolves on its own within 1 to 2 years, some patients may never fully regain shoulder function

Clinical Pearls:

- The pain is typically dull, poorly localized, and may radiate into the biceps
- Attempting to elevate the arm overhead or reach behind one's back may elicit pain and stiffness
- Red flag symptoms: fever, malaise, unintentional weight loss and night sweats
- Cardinal examination findings encompass a reduced range of active and passive motion in all planes (primarily external rotation). In advanced stages, there might be a loss of natural arm swing accompanied by muscular atrophy
- It is diagnosed clinically, with radiography serving to exclude other shoulder pathologies based on history and clinical examination
- Consideration of diabetes and hypothyroidism screening may apply to at-risk patients or those with pertinent signs and symptoms

Management:

- Analgesia
- Physiotherapy: Manual mobilisation techniques
- Intra-articular corticosteroid injection
- Hydrodilatation (arthroscopic distension): a nonoperative procedure that entails injecting a high volume mixture of local anaesthetic, corticosteroids and normal saline into the shoulder capsule to distend and stretch the joint capsule
- Surgery: Manipulation under anaesthesia and arthroscopic capsule release
- Consider referring patients to an orthopedic surgeon if there is minimal symptom improvement after 6 to 12 weeks of conservative management

COMMON UPPER LIMB TENDINOPATHY

Introduction:

- Acute and chronic tendon injuries are prevalent in both athletic, occupational, and lifestyle settings
- Acute tendinopathy (“tendinitis”) usually results from an episode of overuse or strain. There is an acute onset of tendon pain and dysfunction
- Chronic tendinopathy (“tendinosis”) is a degenerative process in which an ineffective healing response results in chronic pain and dysfunction of the affected tendon
- Tendon pathology typically develops in hypovascular or watershed areas, where there is limited blood supply. This leads to reduced oxygenation and a delayed healing process
- The causes of pain in chronic tendinopathy are uncertain but have been attributed to neovascularization, neurochemicals, and mechanical breakdown

Terminology:

- Tendinopathy: Describes the continuum of tendon injuries from an acute to chronic stage
- Tendinitis: An inflammatory response to tendon tissue triggered by an acute injury. Histopathologically, inflammatory cells are present
- Tendinosis: Non-healing and degenerative tendon tissue which is largely devoid of inflammatory cells. Acute tendinitis can occur on a background of chronic tendinosis

Investigations:

- The diagnosis of chronic tendinopathy relies on the patient’s history, tenderness upon palpation, and from physical examination manoeuvres which provoke the tendon (e.g., single leg hop for patellar tendinopathy, single leg heel raise for Achilles tendinopathy)
- Plain radiography may be utilized initially to investigate atraumatic joint pain when considering alternative diagnoses
- Magnetic resonance imaging or ultrasound can be performed if pain persists and initial radiographs are inconclusive or unremarkable

Management:

- The mainstays of treatment:
 - Activity modification with relative rest (i.e., advise patient to move the joint as tolerated – pain score of not more than 3 on the Visual Analogue Scale)
 - Analgesia
 - Physiotherapy for rehabilitative exercises to gradually increase the load-bearing capacity of the injured tendon. This applies to both acute and chronic tendinopathy
- Analgesia:
 - Short-term use of oral or topical NSAIDs or COX-2 inhibitors (if no contraindications) in acute tendinopathy
 - Peritendinous corticosteroid injection with or without fenestration: repeated injections may exacerbate chronic pain and may also lead to tendon rupture. One should never inject corticosteroid into or around a weight bearing tendon (i.e., Achilles tendon)
 - Dry needling
 - Extracorporeal shock wave therapy for treatment-refractory tendinopathy
- Early commencement of targeted and specific rehabilitative exercises is advantageous, as proper loading of the tendon is vital for the healing process
 - Isometric, concentric, and eccentric exercises
- Specialist referral may be considered if conservative therapy proves ineffective for 3 to 6 months

ROTATOR CUFF TENDINOPATHY

- It is the most common cause of shoulder pain
- The rotator cuff comprises of 4 muscles: subscapularis, supraspinatus, infraspinatus, teres minor
- Common presentations include pain and/or weakness with overhead movement, reaching behind one's back, and when lying on the affected side
- Treatment:

- Analgesia
- Rehabilitative exercise: strengthening of the rotator cuff and scapular stabilizers to widen the subacromial space and hence reduce subacromial impingement
- Subacromial corticosteroid injections are commonly utilized for short-term pain relief, particularly valuable when initiating physical therapy
- Specialist referral may be considered if the patient remains symptomatic despite 6 months of physiotherapy

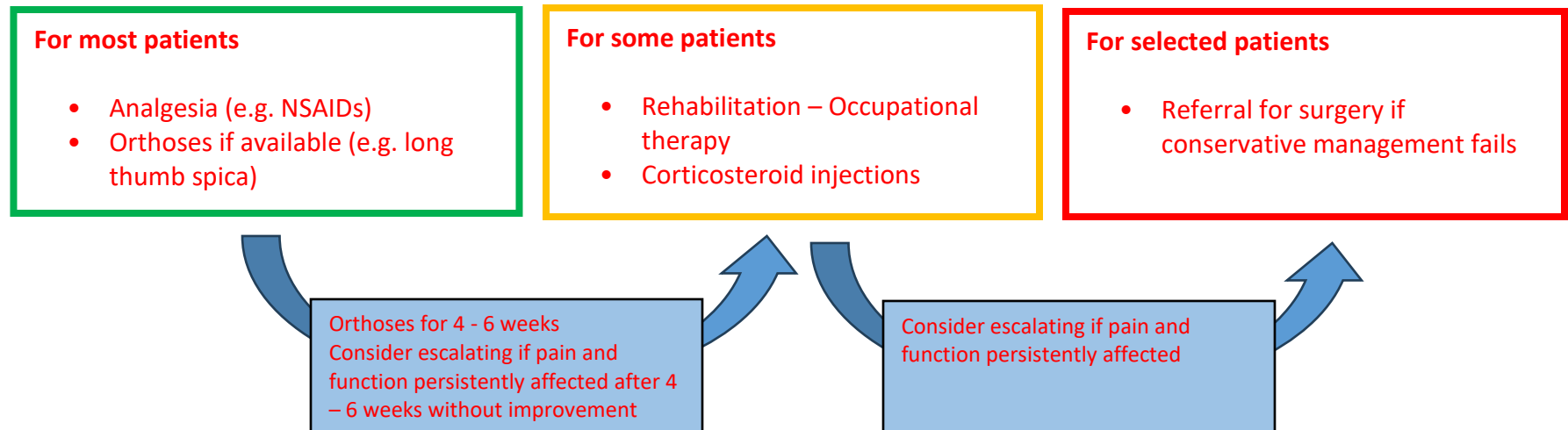
EPICONDYLITIS

- Tendinopathies of the common extensor and flexor tendon origins of the forearm
- Lateral epicondylitis (tennis elbow):
 - Overuse injuries lead to chronic tendinosis of the extensor carpi radialis brevis tendon
 - Patients may exhibit lateral elbow pain during gripping activities and have a reduced grip strength
 - Tenderness on palpation approximately 1 cm distal to the lateral epicondyle, which is exacerbated with resisted wrist extension
 - It may be associated with radial tunnel syndrome (i.e., impingement of the posterior interosseous nerve by the supinator muscle)
- Medial epicondylitis (golfer's elbow):
 - Less common overuse injury leading to chronic tendinosis of the flexor-pronator tendon origin at the medial epicondyle
 - Tenderness around the medial epicondyle exacerbated by resisted forearm pronation and wrist flexion
 - It may be associated with ulnar neuropathy, carpal tunnel syndrome, lateral epicondylitis, and rotator cuff tendinopathy
- Management:
 - Analgesia
 - Rehabilitative exercise: Eccentric strengthening exercises improve the load-bearing capacity of the tendons and relieve pain
 - Bracing: cock-up wrist braces (for tennis elbow) or counterforce straps
 - If conservative measures are ineffective, corticosteroid injection or shockwave therapy may be considered

DE QUERVAIN'S TENOSYNOVITIS

- Tendinitis involving the tendons of the extensor pollicis brevis and abductor pollicis longus in the first extensor compartment of the wrist
- Recurrent inflammation results in stenosing tenosynovitis
- Gradual onset of radial-sided wrist pain worsened by gripping or lifting objects with the wrist in the neutral position
- It commonly affects the dominant wrist, with women affected more frequently than men
- Physical examination
 - Tenderness and swelling over the first dorsal wrist extensor compartment at the radial styloid
 - Provocative test:
 - Finkelstein manoeuvre:
 - Pain over the first extensor compartment (tip of radial styloid) when grasping the patient's thumb and passively adducting the hand ulnarward whilst maintaining longitudinal traction on the thumb
 - More indicative of extensor pollicis longus (EPL) tendon pathology
 - Eichhoff manoeuvre:
 - Pain with ulnar deviation of the wrist while the patient clenches thumb in fist
 - Relief of pain when the thumb is extended while the wrist remains in ulnar deviation
 - Negative grind test of the thumb (positive in the first carpometacarpal osteoarthritis)

- Management
 - A trial of non-surgical conservative management is recommended for most patients
 - A suggested management pathway can be found below



CARPAL TUNNEL SYNDROME

Introduction:

- Common entrapment neuropathy involving compression of the median nerve under the transverse carpal ligament
- Symptoms
 - Characteristic features include night awakening with symptoms and alleviating them by shaking hands (flick sign)
 - Paraesthesia and pain in the median nerve distribution
 - Late-stage symptoms include weakness in thumb abduction and opposition.
- Causes
 - Mostly idiopathic
 - Predisposing conditions include rheumatoid arthritis, diabetes mellitus, pregnancy, hypothyroidism, obesity, acromegaly, and previous wrist fractures
 - Occupational risk factors involve activities characterized by repetitive forceful repetitions or the use of vibratory tools

Physical examination:

- Paraesthesia in the palmar aspect of the thumb, index, middle, and radial half of the ring finger
- Wasting of thenar eminence (severe disease)
- Weakness of thumb abduction and opposition (severe disease)
- Provocative tests: Tinel's test, Phalen's test, Duran's test
- A complete examination of the entire upper extremity (neck, shoulder, elbow, and wrist) should be conducted to exclude other causes

Management:

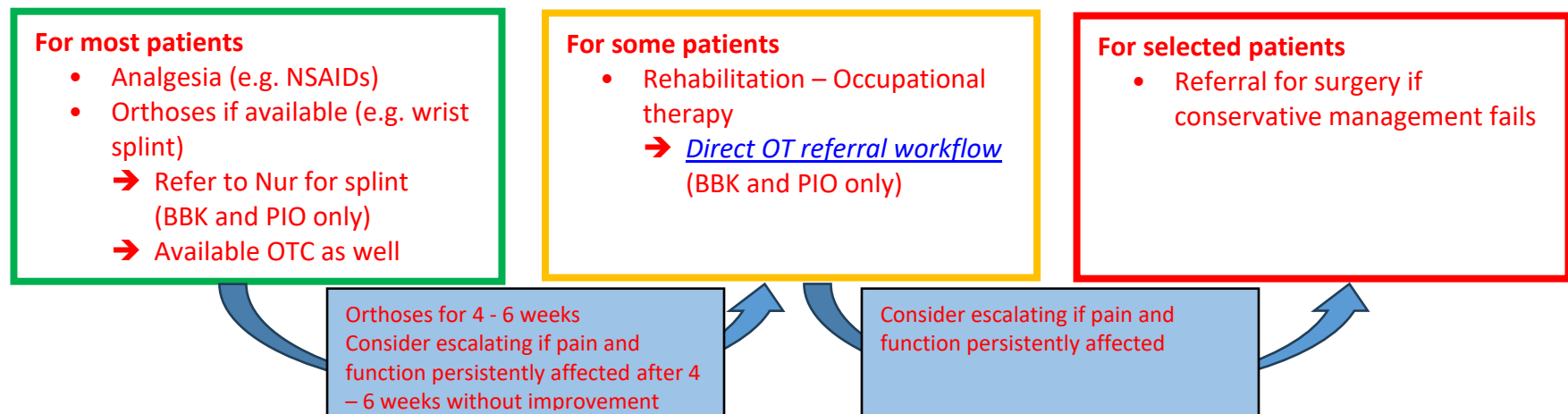
- Conservative
 - Splinting to keep the wrist in a neutral position (usually nocturnal but can be applied continuously)
 - Corticosteroid injection

- Oral prednisolone (20 mg daily for 10 to 14 days) improves symptoms and function but is less effective than corticosteroid injection
- **Occupational** therapy, nerve gliding exercises and activity modification.
- Referral to Hand Surgery
 - Electrodiagnostic tests in atypical cases
 - Surgical decompression for persistent and severe cases

WRIST SPLINT (AVAILABLE IN NUR TREATMENT ROOM FOR BBK AND PIO)

– LINK TO PATIENT INFORMATION LEAFLETS

- Designed to maintain neutral wrist positioning to prevent compression and stretching of median nerve
- It helps to alleviate numbness and pain
- Splint should be always worn daily (even at night) for maximum result
- It can be removed during hand washing, showering, and when performing home exercises for the affected hand
- Recommended wearing time: at least 3 weeks.
- Overall treatment duration: 6 to 8 weeks.
- A suggested management pathway can be found below:



TRIGGER FINGER

Introduction:

- Stenosing tenosynovitis prevents smooth motion of the gliding tendon, usually at the level of A1 pulley
- Prevalent in individuals over 50 years old, particularly among females and those with diabetes
- It commonly affects the thumb, middle and ring fingers.
- Symptoms
 - Pain and stiffness at the volar aspect of the metacarpophalangeal joint (MCPJ)
 - Snapping or popping sensation with digital flexion and extension
 - Locking of the finger in a flexed position at the proximal interphalangeal joint (PIPJ)
- Risk factors include repetitive forceful gripping.

Physical examination:

- Tenderness at the volar aspect of MCPJ directly over the A1 pulley,
- Palpable nodule at the flexor tendon
- Green's Classification
 - Grade I: Pain and history of catching but not demonstrable on physical examination; tenderness over A1 pulley
 - Grade II: Demonstrable catching with intact active digital extension
 - Grade IIIA: Demonstrable catching requiring passive extension of digit
 - Grade IIIB: Demonstrable catching with loss of active flexion
 - Grade IV: fixed flexion contracture of PIPJ

Management:

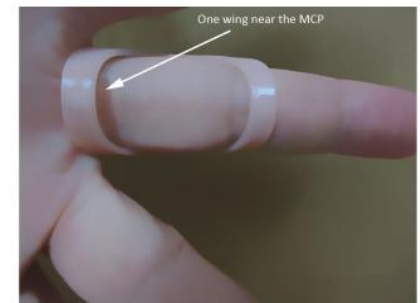
- Conservative
 - Oval 8 splint

- Corticosteroid injection
- NSAIDs
- Occupational therapy and activity modification
- Referral to Hand surgery to consider surgery for persistent cases

OVAL 8 SPLINT (AVAILABLE IN NUR TREATMENT ROOM FOR BBK AND PIO)

– LINK TO PATIENT INFORMATION LEAFLETS

- Designed to limit full finger flexion at the Proximal Interphalangeal (PIP) joint
- It helps reduce stress on inflamed tendons and pulleys
- The splint features wide, smooth bands and three points of control for stability and proper alignment, while allowing functional movements of the fingers
- Available in 14 sizes (sizes 2 through 15).
- Fit can be looser or tighter depending on which end of the splint is placed on the finger first.
- Recommended wearing time: at least 3 weeks.
- Overall treatment duration: 6 to 8 weeks.
- A suggested management pathway can be found below:



For most patients

- Analgesia (e.g. NSAIDs)
- Orthoses if available (e.g. Oval 8 splint)
 - ➔ Refer to Nur for splint (BBK and PIO only)
 - ➔ Available OTC as well

For some patients

- Rehabilitation – Occupational therapy
 - ➔ [Direct OT referral workflow](#) (BBK and PIO only)

For selected patients

- Referral for surgery if conservative management fails

Orthoses for 4 - 6 weeks
Consider escalating if pain and function persistently affected after 4 – 6 weeks without improvement

Consider escalating if pain and function persistently affected

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